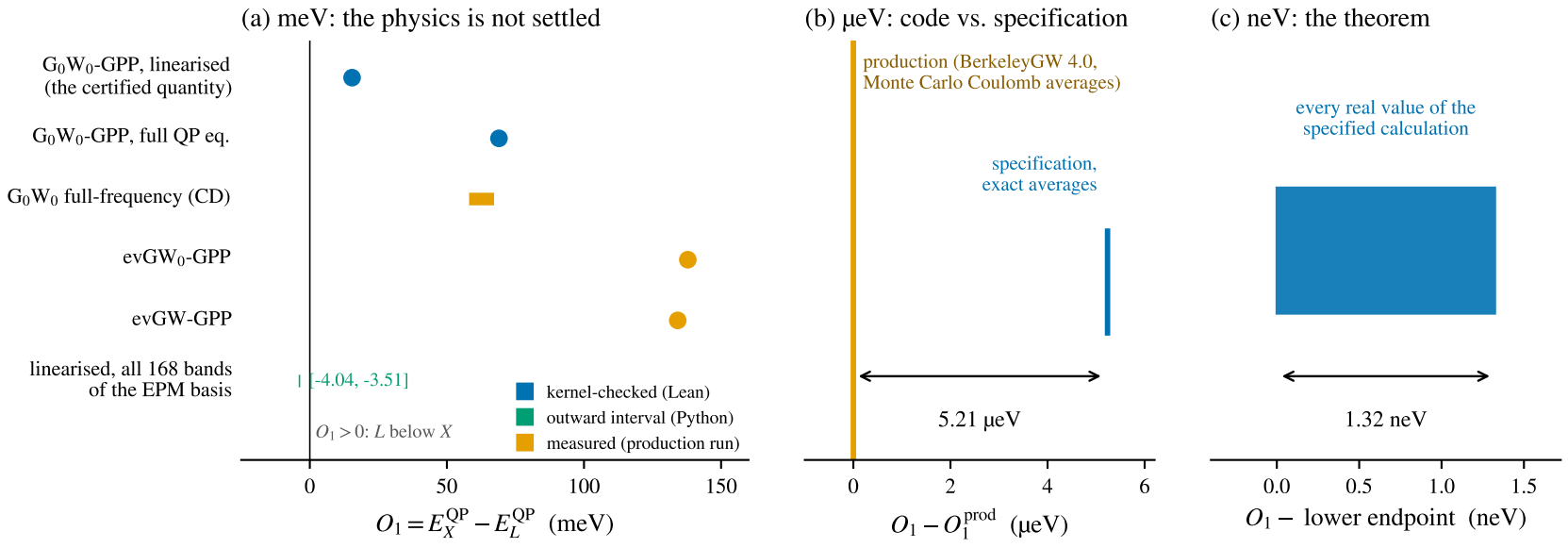
(a) meV: the physics is not settled
G₀W₀-GPP, linearised (the certified quantity)
G₀W₀-GPP, full QP eq.
G₀W₀ full-frequency (CD)
evGW₀-GPP
evGW-GPP
linearised, all 168 bands of the EPM basis
[-4.04, -3.51]
O₁ > 0: L below X
kernel-checked (Lean)
outward interval (Python)
measured (production run)
0
50
100
150
O₁ = E_X^QP − E_L^QP (meV)
(b) μeV: code vs. specification
production (BerkeleyGW 4.0, Monte Carlo Coulomb averages)
specification, exact averages
5.21 μeV
0
2
4
6
O₁ − O₁^prod (μeV)
(c) neV: the theorem
every real value of the specified calculation
1.32 neV
0.0
0.5
1.0
1.5
O₁ − lower endpoint (neV)